

DOKUMENTACJA GAIA

1. Opis i Wymagania

Celem proponowanego systemu jest dynamiczne wspomaganie dojazdu pojazdów uprzywilejowanych (np. karettek, straży pożarnej) w warunkach miejskich, minimalizując czas przejazdu w ruchu drogowym i poprawiając bezpieczeństwo uczestników ruchu. System będzie wykorzystywał technologie agentów, aby w czasie rzeczywistym reagować na zmienne warunki drogowe oraz interakcje między różnymi uczestnikami ruchu.

- System umożliwia rozpoczęcie trasy do punktu docelowego
- Jego pozycja oraz pozycja zwykłych pojazdów jest na bieżąco odczytywana.
- Wydarzenia losowe (wypadki, korki) są monitorowane.
- Trasa jest dostosowywana do warunków.
- Sygnalizacja świetlna jest monitorowana i dostosowuje się do trasy pojazdu.
- Powiadomianie kierowców o zbliżającym się pojeździe:● Przez aplikacje● Dodatkowe wyświetlacze

2. Model Ról

Role: EmergencyVehicle

Description	Autonomiczny agent reprezentujący pojazd uprzywilejowany (np. karetka, samochód straży pożarnej) w systemie. Odpowiada za zarządzanie trasą, odczytywanie pozycji i reagowanie na zmieniające się warunki drogowe.
Responsibilities	Liveness: (InitiateRoutePlanning . RequestPositionUpdate . RequestRouteAdjustment . RequestLightAdjustment)^ω Safety: UniqueID(VehicleID) != Null VehiclePosition != Null RoutePlan != Null EstimatedArrivalTime >= CurrentTime TrafficConditions != Null
Permissions	READ: DestinationPoint, VehiclePosition, OtherVehiclePositions, EventDetails, RoutePlan, AdjustedRoutePlan, TrafficLightID, EstimatedArrivalTime WRITE: RoutePlan, AdjustedRoutePlan, EstimatedArrivalTime CONSUME: RoutePlan, VehiclePosition, OtherVehiclePositions, EventDetails, AdjustedRoutePlan, AdjustedLightTiming PRODUCE: DestinationPoint, VehicleID, CurrentPosition, TrafficConditions, TrafficLightID, EstimatedArrivalTime
Activities	
Protocols	InitiateRoutePlanning, RequestPositionUpdate, RequestRouteAdjustment, RequestLightAdjustment

Role: TrafficMonitor

Description	Agent monitorujący stan ruchu drogowego, w tym pozycje pojazdów, wydarzenia losowe (np. wypadki, korki) oraz sygnalizację świetlną. Zbiera dane z różnych źródeł i przekazuje je do systemu decyzyjnego.
Responsibilities	Liveness: (ReportEvent)^ω Safety: UniqueID(VehicleID) != Null VehiclePosition != Null EventDetails != Null
Permissions	READ: VehicleID, VehiclePosition, OtherVehiclePositions, EventDetails WRITE: EventDetails CONSUME: VehiclePosition, OtherVehiclePositions, EventDetails PRODUCE: VehicleID, VehiclePosition, OtherVehiclePositions, EventDetails
Activities	
Protocols	ReportEvent

Role: RouteOptimizer

Description	Agent odpowiedzialny za dynamiczne dostosowywanie trasy pojazdu uprzywilejowanego w czasie rzeczywistym, w zależności od aktualnych warunków ruchu i dostępnych informacji o przeszkodach.
Responsibilities	Liveness: (InitiateRoutePlanning . ReportEvent . RequestRouteAdjustment)^ω Safety: UniqueID(DestinationPoint) != Null RoutePlan != Null UpdatedRoutePlan != Null AdjustedRoutePlan != Null
Permissions	READ: DestinationPoint, VehiclePosition, OtherVehiclePositions, EventDetails, CurrentPosition, TrafficConditions WRITE: RoutePlan, UpdatedRoutePlan, AdjustedRoutePlan CONSUME: DestinationPoint, VehiclePosition, OtherVehiclePositions, EventDetails, CurrentPosition, TrafficConditions PRODUCE: RoutePlan, UpdatedRoutePlan, AdjustedRoutePlan
Activities	
Protocols	InitiateRoutePlanning, ReportEvent, RequestRouteAdjustment

Role: TrafficLightController

Description	Agent zarządzający sygnalizacją świetlną na skrzyżowaniach. Dostosowuje czasy migania światła w zależności od nadchodzącego pojazdu uprzywilejowanego.
Responsibilities	Liveness: (RequestLightAdjustment) ^ω Safety: UniqueID(TrafficLightID) != Null EstimatedArrivalTime >= CurrentTime AdjustedLightTiming != Null
Permissions	READ: TrafficLightID, EstimatedArrivalTime WRITE: AdjustedLightTiming CONSUME: TrafficLightID, EstimatedArrivalTime PRODUCE: AdjustedLightTiming
Activities	
Protocols	RequestLightAdjustment

Role: DriverNotifier

Description	Agent odpowiedzialny za powiadamianie kierowców o zbliżającym się pojeździe uprzywilejowanym poprzez aplikację i dodatkowe wyświetlacze.
Responsibilities	Liveness: (SendNotification) ^ω Safety: UniqueID(EmergencyVehicleID) != Null EstimatedArrivalTime >= CurrentTime RoutePath != Null NotificationSent != Null
Permissions	READ: EmergencyVehicleID, EstimatedArrivalTime, RoutePath WRITE: NotificationSent CONSUME: EmergencyVehicleID, EstimatedArrivalTime, RoutePath PRODUCE: NotificationSent
Activities	
Protocols	SendNotification

3. Model Interakcji

Global	Name	Initiator	Responder	Inputs	Outputs	Description
EmergencyVehicle requests route planning	InitiateRoutePlanning	EmergencyVehicle	RouteOptimizer	DestinationPoint	RoutePlan	EmergencyVehicle requests RouteOptimizer to plan a route to a specified destination.
EmergencyVehicle requests real-time updates	RequestPositionUpdate	EmergencyVehicle	TrafficMonitor	VehicleID	VehiclePosition, OtherVehiclePositions	EmergencyVehicle requests TrafficMonitor for current position of the vehicle and other road users.
TrafficMonitor reports events to the RouteOptimizer	ReportEvent	TrafficMonitor	RouteOptimizer	EventDetails	UpdatedRoutePlan	TrafficMonitor sends event details (e.g., accidents, traffic jams) to RouteOptimizer for route adjustment.
EmergencyVehicle requests dynamic route adjustment	RequestRouteAdjustment	EmergencyVehicle	RouteOptimizer	CurrentPosition, TrafficConditions	AdjustedRoutePlan	EmergencyVehicle asks RouteOptimizer to adjust the route based on real-time conditions.
EmergencyVehicle requests traffic light adjustment	RequestLightAdjustment	EmergencyVehicle	TrafficLightController	TrafficLightID, EstimatedArrivalTime	AdjustedLightTiming	EmergencyVehicle informs TrafficLightController about its approach to a traffic light.
DriverNotifier sends notifications to drivers	SendNotification	DriverNotifier	DriverNotificationSystem	EmergencyVehicleID, EstimatedArrivalTime, RoutePath	NotificationSent	DriverNotifier notifies drivers via the DriverNotificationSystem about approaching emergency vehicles.

4. Model Usage

Service	Agent	In/Out	Pre/Post Condition	Derived From
GenerateRoutePlan	RouteOptimizationAgent	IN: DestinationPoint OUT: RoutePlan	PRE: EmergencyVehicleAgent has access to DestinationPoint; RouteOptimizationAgent has permission to read DestinationPoint and write RoutePlan POST: RoutePlan is generated and provided to EmergencyVehicleAgent; EmergencyVehicleAgent has permission to consume RoutePlan	InitiateRoutePlanning
RequestPositionUpdate	TrafficMonitoringAgent	IN: VehicleID OUT: VehiclePosition, OtherVehiclePositions	PRE: EmergencyVehicleAgent has VehicleID and requests position update from TrafficMonitoringAgent; TrafficMonitoringAgent has permission to read VehicleID and provide VehiclePosition and OtherVehiclePositions POST: VehiclePosition and OtherVehiclePositions are provided to EmergencyVehicleAgent; EmergencyVehicleAgent has permission to consume VehiclePosition and OtherVehiclePositions	RequestPositionUpdate
ProvidePositionData	EmergencyVehicleAgent	IN: VehicleID OUT: VehiclePosition, OtherVehiclePositions	PRE: EmergencyVehicleAgent has VehicleID and can provide position data to TrafficMonitoringAgent; TrafficMonitoringAgent has permission to consume VehiclePosition and OtherVehiclePositions POST: VehiclePosition and OtherVehiclePositions are provided to TrafficMonitoringAgent; TrafficMonitoringAgent has permission to write VehiclePosition and OtherVehiclePositions	RequestPositionUpdate
RequestRouteAdjustment	EmergencyVehicleAgent	IN: CurrentPosition, TrafficConditions OUT: AdjustedRoutePlan	PRE: EmergencyVehicleAgent has CurrentPosition and TrafficConditions; RouteOptimizationAgent has permission to consume CurrentPosition and TrafficConditions; TrafficMonitoringAgent has provided CurrentPosition and TrafficConditions to EmergencyVehicleAgent POST: AdjustedRoutePlan is generated and sent to RouteOptimizationAgent; RouteOptimizationAgent has permission to write AdjustedRoutePlan	RequestRouteAdjustment
RequestLightAdjustment	EmergencyVehicleAgent	IN: TrafficLightID, EstimatedArrivalTime OUT: AdjustedLightTiming	PRE: EmergencyVehicleAgent has TrafficLightID and EstimatedArrivalTime; TrafficLightControlAgent has permission to consume TrafficLightID and EstimatedArrivalTime POST: AdjustedLightTiming is provided to TrafficLightControlAgent; TrafficLightControlAgent has permission to write AdjustedLightTiming	RequestLightAdjustment

Service	Agent	In/Out	Pre/Post Condition	Derived From
ProcessEventDetails	RouteOptimizationAgent	IN: EventDetails OUT: UpdatedRoutePlan	PRE: TrafficMonitoringAgent has EventDetails and reports to RouteOptimizationAgent; RouteOptimizationAgent has permission to read EventDetails and write UpdatedRoutePlan POST: UpdatedRoutePlan is generated and sent to EmergencyVehicleAgent; EmergencyVehicleAgent has permission to consume UpdatedRoutePlan	ReportEvent
SendNotification	DriverNotificationAgent	IN: EmergencyVehicleID, EstimatedArrivalTime, RoutePath OUT: NotificationSent	PRE: DriverNotificationAgent has EmergencyVehicleID, EstimatedArrivalTime, and RoutePath; DriverNotificationAgent has permission to read EmergencyVehicleID, EstimatedArrivalTime, and RoutePath POST: NotificationSent is confirmed and delivered to DriverNotificationSystem; DriverNotificationAgent has permission to write NotificationSent	SendNotification

5. Model Znajomości

